

5. The study of bones is done under the branch of science called -

- (a) Geology (b) Serology
(c) Orology (d) Osteology
(e) None of the above/More than one of the above

65th B.P.S.C. (Pre) 2019

Ans. (d)

Osteology	—	Study of bones
Orology	—	Study of mountains
Serology	—	Study of serum
Geology	—	Study of internal structure of the earth and matter found inside it

6. Ornithology is the study of

- (a) dogs (b) bones
(c) birds (d) snakes

B.P.S.C. Assistant Mains (GK) Paper II 2023

Ans. (c)

Ornithology is the study of Birds. Study of snakes (Reptiles) and Frog (Amphibia) comes under the Herpetology. Study of the structure and function of Bones comes under Osteology.

Evolution of Life

1. The main factor of development is -

- (a) Mutation (b) Acquired qualities
(c) Sexual reproduction (d) Natural selection

39th B.P.S.C. (Pre) 1994

Ans. (d)

Today the most accepted theory of development is known as the synthesis theory of development. In this, the origin of species has been explained on the basis of interaction of genetic variations and natural selection.

2. According to recent proof, the life on earth emerged approximately -

- (a) Before 20,000 years
(b) Before 2,00,000 years
(c) Before 20,00000 years
(d) Before 2,000,000,000 years

40th B.P.S.C. (Pre) 1995

Ans. (d)

The age of the earth is about 4.6 billion years. This dating is based on evidence from radiometric age dating of meteorite materials and is consistent with the radiometric ages of the oldest-known terrestrial and lunar samples. According to recent proof, the life on earth emerged about 2,000,000,000 years ago.

3. The book, *The Origin of Species* was written by -

- (a) Linnaeus (b) Lamarck
(c) Mendel (d) Darwin
(e) None of the above/More than one of the above

65th B.P.S.C. (Pre) 2019

Ans. (d)

Charles Darwin explained natural selection in his book 'The Origin of Species' (1859; Full Title - On the Origin of Species by Means of Natural Selection, or the Preservation of Favoured Races in the Struggle for Life). It is considered to be the foundation of evolutionary biology.

4. Who explained first the evolution of life?

- (a) Newton (b) Einstein
(c) Charles Darwin (d) Lamarck

43rd B.P.S.C. (Pre) 1999

Ans. (d)

French biologist Lamarck is best known for his "Theory of Inheritance of Acquired Characters", first presented in 1801. He was the first person to explain the theory of evolution to the public. In 1809, this theory was published in his book 'Philosophie Zoologique'.

5. The reason of large-scale diversity among the organisms is -

- (a) Adaptation (b) Cooperation
(c) Mutation (d) Polyploidy

56th to 59th B.P.S.C. (Pre) 2015

Ans. (c)

Three basic and accessory factors have been recognized to provide genetic variability to the gene pool of a population. Amongst the basic factors, gene and chromosomal mutations have been recognized as the ultimate sources of biological changes and hence, the ultimate 'raw material' of evolution. Gene recombinations, the third basic factor then increases the variability of individual genotypes in a population.

6. The reason for development is -

- (a) Fossil material
(b) Struggle to live
(c) Generation to generation adaptation
(d) Inheritance

46th B.P.S.C. (Pre) 2004

Ans. (c)

The most accepted theory of development is known as the synthesis theory of development. In this, the emergence of species can be understood on the basis of interactions of genetic variations and natural selection. That is, adaptation from generation to generation is the reason for development.

7. Latest ancestor of modern human is

- (a) Java man (b) Cro-Magnon man
(c) Neanderthal man (d) Peking man

39th B.P.S.C. (Pre) 1994

Ans. (b)

Cro-Magnon is a common name that has been used to describe the first early modern humans. The earliest known remains of Cro-Magnon like humans are radiocarbon dated to near about 35,000 to 40,000 years before. Cro-Magnons were robustly built and powerful. Their body was generally heavy and solid with a strong musculature. The face was short and wide. The chin was prominent. The brain capacity was about 1,600 cubic centimeters, larger than the average for modern humans. However, recent research suggests that the physical dimensions of so-called 'Cro-Magnon' are not sufficiently different from modern humans to warrant a separate designation.

8. Archaeopteryx is –

- (a) Most ancient bird of Jurassic age
(b) Reptile of Jurassic period
(c) Reptile of Triassic period
(d) Reptile of both Triassic and Jurassic period

40th B.P.S.C. (Pre) 1995

Ans. (a)

Archaeopteryx is a connecting link between reptilia and aves. It is a creature in which we found the character of both reptiles and birds. Archaeopteryx dates from about 150 million years ago during the Jurassic period when many dinosaurs lived. It is one of the oldest-known birds. It is the first bird and modified from reptiles.

9. Which of the following is not a vestigial organ in humans?

- (a) Nictitating membrane
(b) Auricular muscles of pinna
(c) Front flattened teeth (Incisors)
(d) Vermiform Appendix

46th B.P.S.C. (Pre) 2004

Ans. (c)

The morphological and anatomical studies in living organisms give us knowledge about the residual organs. It is such an organ, which appears to be non-functional, but it can be shown to be homologous to the functional part of the ancestor. For example, the vermiform Appendix present in the large intestine of humans, the nictitating membrane and auricular muscle of pinna are active in mammals.

10. The exact place where the living being lives is called -

- (a) Habitat (b) Ecosystem

- (c) Niche (d) biome
(e) none of the above / more than one of the above

65th B.P.S.C. (Pre) 2019

Ans. (a)

The exact place of living of plants or animals or other organisms of a particular species is called Habitat.

11. Which of the following are constituents of an ecosystem?

- (a) Animals (b) Plants
(c) Microorganisms (d) More than one of the above
(e) None of the above

B.P.S.C. PRT 2023 Shift I

Ans. (d)

An ecosystem is a community of organisms and their physical environment interacting together. Environment involves both living organisms and the non-living physical conditions. These two are inseparable but inter-related.

12. The flow of energy in an ecosystem is

- (a) multidirectional (b) unidirectional
(c) tridirectional (d) More than one of the above
(e) None of the above

B.P.S.C. PRT 2023 Shift I

Ans. (b)

The energy flow of an ecosystem is a fundamental concept of ecological studies. The energy flow of an ecosystem is unidirectional.

13. Himalayan Brown bear, wild Asiatic buffalo, desert fox are examples of

- (a) rare species (b) extinct species
(c) endangered species (d) More than one of the above
(e) None of the above

B.P.S.C. School Teacher (Class 6-8) 10-12-2023

Ans. (a)

Himalayan Brown bear, wild Asiatic buffalo, desert fox are examples of rare species.

A rare species is a group of organisms that are very uncommon, scarce, or infrequently encountered whereas "Endangered" means a species is in danger of extinction throughout all or a significant portion of its range. "Threatened" means a species is likely to become endangered within the foreseeable future.

14. The translation termination triplet is

- (a) UAA (b) UAU
(c) UGC (d) More than one of the above
(e) None of the above

B.P.S.C. School Teacher (Class 9-10) 07-12-2023

Ans. (a)

The translation termination triplet is UAA. UAG, UGA. These codon are also known as stop codons because the process of protein synthesis is finally stopped at this stage.